

Exponential growth and decay



1

- a
- i Exponential growth
 - ii 0.6
- b
- i Exponential decay
 - ii 0.8

2 233

3 $f(t) = 26030 \times (1.011)^t$

4 $P(t) = 10\,637 (1.0040)^t$

5

a $y = 750 (1.03)^t$

b 869

6

a $y = 2900 (1.09)^v$

b \$5778.43

7

a $y = 322\,000 (0.95)^m$

b \$249 157.46

8

a $y = 2310 (0.974)^n$

b 1328

9

a $y = 200 (3)^n$

b \$48 600

10

a $y = 21 (0.5)^n$

b 0.66 m

11

a $A = 300 (0.5)^{\frac{t}{20}}$

b 0.59 grams

12

- a Yes b No c No d Yes

13 $r = 3.53\%$



14 631 bacteria

15

a 0.01

b $A = 38\,200 \times 0.99^n$

c \$35\,964.54

d Motorbike

16 41\,199

17 6591

18 262

19

a

Quarter(N)	1	2	3	4	5	6	7
Online Customers (C)	480	2880	17280	103\,680	622\,080	3\,732\,480	22\,394\,880

b $C = 480 \times 6^{N-1}$

c 29\,023\,764\,480 sales

d To not assume that growth will change in exactly the same way in the coming quarters and try a different model.